

Campus Recruitment Training – Phase 2

Java Cluster Syllabus proposal

Prerequisites

Students should have basic knowledge of Core Java, OOPs, Collections Framework, Exception Handling, DSA Basics, Time and Space Complexity, SQL and DBMS Basics, ER Diagrams, HTML, CSS, JavaScript, REST APIs, JSON, Client-Server Architecture, Git and GitHub, Postman, Maven, Basic Linux Commands

Phase 1: Advanced Backend with Spring Boot (20 Hours)

- **Foundation & Request Lifecycle:** Transitioning from JDBC/Servlets to Spring Boot. Understanding the **DispatcherServlet**, Inversion of Control (IoC), and the Application Context.
- **Spring Boot Internal Architecture:** Understanding the "Magic" of Spring Boot. Deep dive into Annotations (@RestController, @Service, @Repository), Dependency Injection, and how the Application Context works.
- **Data Access Layer (JPA/Hibernate):** Moving beyond raw SQL. Teaching students how to map Java Objects to Database Tables using **Spring Data JPA**. Emphasis on writing custom JPQL queries and handling "One-to-Many" or "Many-to-Many" relationships.
- **Secure API Design:** Implementing **JWT (JSON Web Tokens)** for secure authentication. Students will learn how to protect specific "admin" routes and handle user logins-a standard requirement for companies like **GlobalLogic** and **Accenture**.
- **Assignment Project :** Students will develop a Secure Employee Management REST API with JWT Authentication, role-based access control, CRUD operations, database integration using Spring Data JPA, and API validation/exception handling. The project will simulate real enterprise backend development workflows using Spring Boot and MySQL/PostgreSQL.

Phase 2: Modern Frontend Integration (15 Hours)

Focus: Building a sleek, professional user interface that talks to the Java backend.



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- **Asynchronous Communication:** Using **Axios** or **Fetch API** to send requests to the Spring Boot server. Handling "Loading" states, "Error" messages, and displaying dynamic data in tables and cards.
- **Modern Styling & UX:** Using frameworks like **Tailwind CSS** or **Bootstrap** to ensure the projects look like professional products, which significantly impresses interviewers during demo rounds.
- **Assignment Project :** Students will build a Full Stack Admin Dashboard by integrating a React.js frontend with the Spring Boot backend. The project will include authentication handling, API integration using Axios/Fetch API, dynamic dashboards, CRUD operations, responsive UI design, and frontend state management.

Phase 3: Microservices & System Design (15 Hours)

Focus: High-level architectural thinking required for 10+ LPA roles like Nutanix and Porter.

- **Breaking the Monolith:** Teaching students why and how to split one big application into smaller, independent services (e.g., an "Order Service" and a "User Service"). Including practical scaling numericals such as estimating service load, request distribution, concurrent users, and database growth after service separation.
- **Service Orchestration:** Hands-on with **Netflix Eureka** for Service Discovery (how services find each other) and **API Gateways** (the single entry point for the frontend). Including numericals related to API Gateway throughput, request routing, latency calculation, and load balancing across multiple service instances.
- **Assignment Project :** Students will implement an E-Commerce Microservices System consisting of independent services such as User Service, Product Service, Order Service, and Payment Service. The project will include Service Discovery, API Gateway, Circuit Breakers, load balancing, and scaling/failure-handling numericals using Spring Cloud and Resilience4j.

Phase 4: Quality, Deployment & Portfolio (10 Hours)

Focus: The "Finishing School" aspect—making the code production-ready.

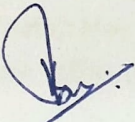
- **The Art of Testing:** Writing **Unit Tests** using **JUnit 5** and **Mockito**. Companies like **EPAM** value "Test-Driven Development" (TDD) and will often ask candidates to write a test case during the interview.
- **Containerization with Docker:** Learning to "Dockerize" the application. This ensures the app runs exactly the same on the student's laptop as it does on the recruiter's machine.
- **Assignment Project :** Students will make their application production-ready by implementing unit testing using JUnit5 and Mockito, containerization using Docker,


(Kaushik Ray)

CI/CD basics using GitHub Actions, and cloud deployment fundamentals. The final outcome will be a deployable enterprise-grade project suitable for placement portfolios and technical interviews.

Companies Mapped by Package Range

Package Range	Companies
20 LPA and above	Nutanix, DE Shaw & Co
15–20 LPA	ZS, Porter, Kickdrum
10–15 LPA	Amadeus, Schneider Electric, Aspect Ratio, PubMatic, Varroc
8–10 LPA	Infosys Specialist Programmer, HSBC, EPAM, Principal Global Services, C&R Software, Planful
6–8 LPA	Finastra, Fendahl, Cognida.ai, Vconstruct, Softlink International, Accenture Advanced Role
4–6 LPA	GlobalLogic, Cognizant GenC Next, Capgemini, Bristlecone, Sagacious IP
Below 4 LPA	Infosys System Engineer, Cognizant GenC, Turing Business Analyst


(Kaushik Ray)